Linking Portuguese Parliamentary Data: Integrating Biographical and Electoral Information

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Resumo

The Assembleia da República (AR) publishes multiple open XML datasets describing its institutional structure and the biographical profiles of Members of Parliament. However, these datasets are heterogeneous and not directly interoperable. This work presents a domain-specific RDF ontology and data transformation pipeline that converts and integrates parliamentary structure and biographical data into a validated knowledge graph. While inspired by existing efforts such as the Map4Scrutiny ontology [1], the proposed model focuses on institutional and biographical aspects of parliamentary activity that are outside the scope of interest-declaration and procurement analysis. The resulting knowledge graph enables consistent querying and analysis of parliamentary data across multiple legislatures.

1 Introduction

Semantic modeling has been increasingly adopted to improve the interoperability and analytical value of parliamentary open data. The Map4Scrutiny (M4S) ontology [1] provides a reference framework for representing relationships between Members of Parliament, private interests, and public procurement, supporting transparency and accountability analyses.

In parallel, the Portuguese Parliament publishes extensive open data [1] describing its internal organization and representatives, namely through the *Informação Base* and *Registo Biográfico* datasets. Although rich, these datasets are published in XML and lack a unified semantic structure, limiting their combined use.

This work addresses this gap by defining an independent ontology and RDF conversion pipeline to model legislatures, parliamentary roles, party affiliations, and biographical information in a coherent and machine-interpretable form.

2 Requirements

The project was required to implement an end-to-end semantic data pipeline over official data from the AR. The concrete requirements were as follows:

- **Data extraction**: The system had to retrieve XML data from the Portuguese Parliament open data portal, covering both *Informação Base* (parliamentary structure and mandates) and *Registo Biográfico* (biographical information), for each legislature.
- **XML processing**: Extracted XML needed to be correctly parsed and cleaned, including handling encoding issues such as the presence of a Byte Order Mark (BOM).
- **Ontology Definition**: An ontology was created from scratch, using *Protégé*, to model several entities such as *Member of Parliament* (MoP), *Habilitation*, *Academic Title*, etc.
- **RDF transformation**: The extracted data had to be converted into RDF using the custom ontology, ensuring that all relevant parliamentary entities (e.g., Members of Parliament, legislatures, parliamentary groups, electoral circles) were explicitly represented.

[1] See Dados Abertos AR

- **Biographical data integration**: Biographical information had to be used to enrich existing parliamentary entities, without creating duplicate data when processing cumulative biographical sources.

- **Stable entity identification**: The same real-world entity had to be consistently identified across different data sources, requiring deterministic and uniform URI generation.

- **Semantic validation**: The resulting RDF graph had to be validated using SHACL constraints to ensure structural correctness, valid cardinalities, and overall data consistency.

- **Standards alignment**: The ontology and data needed to align with established vocabularies and ontologies (e.g., Schema.org, SKOS, OCD) to promote interoperability.

- **RDF serialization and querying**: The final knowledge graph had to be serialized in Turtle format and be suitable for querying using SPARQL.

3 Existing Solutions

The Assemblee Data Portal (France), UK Parliament Data Service, and European Parliament Linked Open Data portal already expose parliamentary records in RDF, often using the Parliament Ontology [2] or Popolo [3] data model.

The Parliament Ontology modeled the UK Legislative System, with appropriate classes and attributes for their specific system.

Popolo proposed a more general-use approach and is a W3C community standard and models persons, organizations, memberships, and areas, which are very relevant for parliamentary data. Their approach, on the other hand, is more general-use than others as they didn't seek to represent a specific legislative system, but offer an ontology that could be used by many.

The Italian parliament also has a dedicated ontology called Ontology of the Chamber of Deputies (OCD) [4] where they modeled the entire Italian legislative system.

These projects demonstrate good Linked Data practices, such as the use of persistent URIs, the reuse of vocabularies, and SPARQL endpoints. They can serve as structural and architectural references.

The M4S (Map4Scrutinity) modeled an ontology in order to expose relationships between Members of the Portuguese Parliament, their declared private interests, and public procurement records available through BASE [5]. To do this, they modeled one class (Parlamentarian), some properties such as constituency, parliamentary term, and company shares, and some concept schemes such as parliamentary commissions. However, M4S focuses more on public contracts and procurement. Additionally, because M4S does not provide the AR information it scraped in bulk, expanding its ontology was difficult, since it meant that we had to scrape every information ourselves, including BASE's public contacts, which are especially bothersome. Therefore, we decided to create our own ontology, focused on biographical information about MoP's and electoral circle information.

4 Architecture

1 Shows the system architecture. The main information provided by the **Assembleia da República** (AR) through their open data portal [6]. After fetching the information through a simple HTTP GET request, we had access to the data in XML format. The next step was to convert the XML data into an RDF format, in this case we used Turtle. This format had to comply with the custom ontology we designed. This step also covers the linking process, since everything was done in the same script with 'rdflib'. Finally, this data can now be queried through a SPARQL query.

[2]See UK Parliament Ontology - accessed in December 2025

[3]See Popolo - accessed in December 2025

[4]See OCD - accessed in December 2025

[5]See BASE - accessed in December 2025

[6]See See Dados Abertos - accessed in December 2025

2 Shows the ontology created for the purpose of this project. Some classes modeled by this ontology are: *Member of Parliament (MoP)*, *Habilitation*, *Legislature*, *ElectoralCircle*, etc. To maximize interoperability, we researched various existing vocabularies to reuse terms defined by them, e.g, *dcterms:identifier* for IDs, *schema:jobTitle* for the MoP's profession, *schema:startDate* and *schema:endDate* for dates, and various other terms.

5 Method

5.1 Data acquisition and structuring

While the open data the AR provides is very extensive, encompassing various topics such as *Iniciativas* and *Orçamento do Estado* across every legislature, we decided to focus on 2 datasets in specific: *Informação Base* and *Registos Biográficos*. To reduce processing and querying time, we also just used the two most recent legislatures of each of these datasets.

Informação Base consists of general information about a given legislature, such as start and end date, but also more specific information about the electoral circles, members of parliament, parliamentary groups and legislative sessions. They also provide a small document explaining the tags [7].

The *Registos Biográficos* is a little more complex. This dataset contains biographical information about an MoP such as their *gender*, *academic titles*, *literary habilitations*, *roles* inside the AR, *comittees/unions* they take part in, and many more. Like *Informação Base*, the AR provides a PDF for understanding the XML structure and the meaning of each tag [8].

In order to retrieve the data and prepare it to transform into RDF format we used the *xml.etree* library for Python, which transformed the *XML* file into tree format ready to be used.

[7]See Informação Base PDF - accessed in December 2025

[8]See Registo Biográfico PDF - accessed in December 2025

5.2 Ontology design and extension

We started by modeling what we needed based on the documents provided by the AR. To do this, we developed a class diagram, allowing us to easily visualize it, as seen in 2.

M4S modeled a few classes such as Parlamentarian, some properties such as constituency, parliamentary term and company shares and some concept schemes, such as parliamentary commissions. Comparing with our needs, their approach was too simple, using RDF properties where we felt it made more sense to use OWL classes. Despite this, since their Parlamentarian class still represents the same as our MoP (Member of Parliament) class, we classified them as the same using *owl:equivalentClass*.

Ontologia della Camera dei deputati (OCD) offered many common classes to our needs, such as parliamentary groups and members of parliament. On the other hand, when trying to understand the properties we were met with some Italian-only descriptions, which really made it difficult to understand, hence we opted to not fully use this vocabulary, but instead declare some classes as equivalent, such as *MoP* to *deputato* and *Parliamentary Group* to *gruppo parlamentare*.

The Parliament Ontology on the other hand was too specific for the U.K. system. Since their system works in a way that is not that similar to ours, we decided to discard the reuse of classes and properties, to avoid false equivalences.

Popolo was considered, but ultimately discarded, as their approach was too general for our use case, where we wanted to specifically model the Portuguese parliament.

Afterwards, we looked at what **M4S** did using *Protégé* and by inspecting their *.ttl* file. Although not much was done regarding the ontology, it helped us define some classes in the beginning.

Finally, to maximize interoperability, we created a Vocabulary Terms Matrix [9], with different terms from various existing vocabularies, such as *Schema*, *DCTerms*, *OCD*, etc, with the goal of choosing terms that are aligned with the entities

[9]Vocabulary Terms Matrix

we tried to model and minimize the creation of new terms.

5.3 RDF conversion and linking

To convert the XML format to RDF, we used the *rdflib* Python library.

As mentioned previously in 1, the conversion and linking processes were performed in tandem, since everything was done in the same script, instead of linking using *OpenRefine*. This approach allowed entity creation and enrichment to occur incrementally.

The electoral circles were each linked to their respective *Wikidata* counterparts, while the MoP's were linked to their respective webpage within the AR website.

Finally, to validate the compliance of the final *.ttl* file with our ontology, we created a *SHACL* file.

5.4 Querying

With the RDF graph finished, we created some *SPARQL* queries to test its effectiveness, first starting with an idea expressed in natural language, and only then translating it to *SPARQL* for querying:

1. How many MoP's with a sociology habilitation has each parliamentary group had?
2. How many academic titles are there, in each legislature?
3. Relevant information about electoral circles with >1M electorate (according to *WikiData*), during the XVII legislature.
4. Per legislature, how many MoPs per parliamentary group, whose most recent situation was as Permanent, have a law habilitation?
5. Per legislature, which MoPs from which parliamentary group, started as Permanent but aren't permanent as the latest situation and what jobs do they have?
6. Gender ratio per parliamentary group.

The two most interest to point out here are number 3, due to its complexity and linkage to external sources (*WikiData*), and number 6 due to its results. The results of those queries can be seen in 3 and 4

6 Results

After running the full pipeline for legislatures XVI and XVII, we obtained a single RDF graph containing the institutional and biographical data published by the AR. The final dataset contains 65288 triples and follows the structure defined by the custom ontology.

The generated graph was validated using the SHACL shapes defined for the project, and no violations were reported. This indicates that the conversion process produced data that respects the expected structure and cardinalities. With this,

The SPARQL queries defined in the previous section were executed over the resulting graph and returned the expected results. Queries combining internal parliamentary data with external sources, such as the electorate size of electoral circles retrieved from Wikidata, were successfully executed. The gender distribution query also produced clear and interpretable results per parliamentary group, showing that the modeled data is suitable for analytical queries that were not directly supported by the original XML sources.

7 Conclusion

In this project, we converted parliamentary open data published by the AR, specifically *Informação Base* and *Registos Biográficos* from XML into RDF and integrated it using a custom ontology.

Overall, the project met its initial goals and provides a solid basis for future extensions, such as adding more legislatures, improving the modeling of biographical information, or publishing the data through a SPARQL endpoint.

8 Annexes

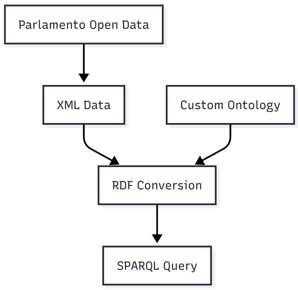


Figura 1: *System Architecture*

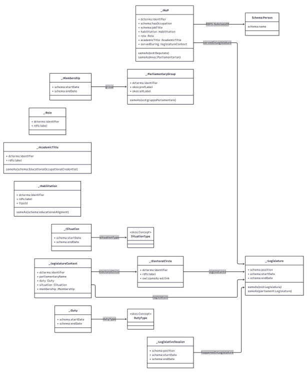
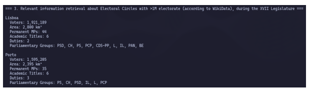


Figura 2: *Custom Ontology defined for the Portuguese Parliament Domain*



```
=== 3. Relevant information retrieval about Electoral Circles with >1M electorate (according to WikiData), during the XVII Legislature ===

Lisboa
  Voters: 1,921,189
  Area: 2,880 km²
  Permanent MPs: 44
  Academic Titles: 6
  Duties: 2
  Parliamentary Groups: PSD, CH, PS, PCP, CDS-PP, L, IL, PAN, BE

Porto
  Voters: 1,595,285
  Area: 2,395 km²
  Permanent MPs: 35
  Academic Titles: 6
  Duties: 3
  Parliamentary Groups: PS, CH, PSD, IL, L, PCP
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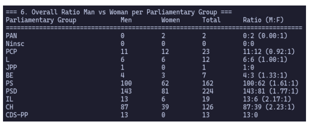
Figura 3: *Result of query number 3*



=== 6. Overall Ratio Man vs Woman per Parliamentary Group ===

Parliamentary Group	Men	Women	Total	Ratio (M:F)
PAN	0	2	2	0:2 (0.00:1)
Ninsc	0	0	0	0:0
PCP	11	12	23	11:12 (0.92:1)
L	6	6	12	6:6 (1.00:1)
JPP	1	0	1	1:0
BE	4	3	7	4:3 (1.33:1)
PS	100	62	162	100:62 (1.61:1)
PSD	143	81	224	143:81 (1.77:1)
IL	13	6	19	13:6 (2.17:1)
CH	87	39	126	87:39 (2.23:1)
CDS-PP	13	0	13	13:0

Figura 4: *Result of query number 6*

Referências

[1] Lopes Inês, Batista Ana Alice, e Afonso Óscar Atanázio. Map4Scrutiny – A Linked Open Data Solution for Politicians Interest Registers. Em *Proceedings of the International Conference on Dublin Core and Metadata Applications*. Dublin Core Metadata Initiative, março 2023.